

Marie-Christine Düker

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AFFILIATION

Assistant Professor, <i>Technical University of Munich</i>	2026–present
Assistant Professor, <i>Friedrich-Alexander University Erlangen-Nürnberg</i>	2023–2026

EDUCATION

<i>Cornell University</i> , Department of Statistics and Data Science Postdoctoral Associate Advisors: Dr. David Matteson	2020–2023
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<i>Ruhr-University Bochum</i> , Faculty of Mathematics Ph.D. Mathematics <i>Summa Cum Laude</i> Thesis: <i>High-dimensional time series under long-range dependence and nonstationarity</i> Advisor: Dr. Herold Dehling	2016–2020
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<i>The University of North Carolina at Chapel Hill</i> , Department of Statistics and Operations Research Visiting Scholar Advisor: Dr. Vladas Pipiras	2018–2019
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<i>Ruhr-University Bochum</i> , Faculty of Mathematics MS Mathematics with Minor Economics <i>Summa Cum Laude</i>	2014–2016
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<i>Ruhr-University Bochum</i> , Faculty of Mathematics BS Mathematics with Minor Economics	2011–2014
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PUBLICATIONS

PEER-REVIEWED PAPERS

- [1] M.-C. Düker and S. Bai. On extremes for Gaussian subordination. *Stochastic Processes and their Applications*, 2026 [\[arXiv\]](#).
- [2] Y. Kim, M.-C. Düker, Z. F. Fisher and V. Pipiras. Latent Gaussian dynamic factor modeling and forecasting for high-dimensional count time series. *Journal of Time Series Analysis*, 2025: [\[Link\]](#) [\[arXiv\]](#).
- [3] A. Betken and M.-C. Düker. Second order asymptotics of the empirical process under long-range dependence. *Econometric Theory*, 2025: [\[Link\]](#) [\[arXiv\]](#).
- [4] M.-C. Düker, D. S. Matteson, T. Ruey and I. Wilms. Vector AutoRegressive Moving Average Models: A Review. *WIREs Computational Statistics*, 2025 [\[Link\]](#) [\[arXiv\]](#).

- [5] M.-C. Düker, R. Lund and V. Pipiras. High-dimensional latent Gaussian count time series: Concentration results for autocovariances and applications. *Electronic Journal of Statistics*, 18(2):5484–5562, 2024: [\[doi\]](#) [\[arXiv\]](#).
- [6] Y. Xu, M.-C. Düker, and D. S. Matteson. Testing simultaneous diagonalizability. *Journal of the American Statistical Association*, 119(546):2386–2414, 2024: [\[doi\]](#) [\[arXiv\]](#).
- [7] C. Baek, M.-C. Düker, and V. Pipiras. Local Whittle estimation of high-dimensional long-run variance and precision matrices. *The Annals of Statistics*, 51(6):2386–2414, 2023: [\[doi\]](#) [\[arXiv\]](#) [\[Supplement\]](#).
- [8] M.-C. Düker, S.-O. Jeong, and T. Lee, C. Baek. Detection of multiple change-points in high-dimensional panel data with cross-sectional and temporal dependence. *Statistical Papers*, 1–33, 2023: [\[doi\]](#) [\[Link\]](#).
- [9] M. Davidow, T. Schafer, M. Cory, J. Che-Castaldo, M.-C. Düker, E. Feng, D. S. Matteson. Clustering Future Scenarios Based on Predicted Range Maps. *Methods in Ecology and Evolution*, 14(5), 1346–1360, 2023: [\[doi\]](#) [\[arXiv\]](#).
- [10] C. Goolsby, J. Losey, A. Fakharzadeh, Y. Xu, M.-C. Düker, M. Sherman Getmansky, D. S. Matteson, M. Moradi. Addressing the embeddability problem in transition rate estimation from Markov state models. *Journal of Physical Chemistry A*, 127(27), 5745–5759, 2023: [\[doi\]](#) [\[bioarXiv\]](#).
- [11] M.-C. Düker, V. Pipiras, and R. R. Sundararajan. Cotrending: testing for common deterministic trends in varying means model. *Journal of Multivariate Analysis*, page 104825, 2021: [\[doi\]](#) [\[Link\]](#) [\[Supplement\]](#).
- [12] M.-C. Düker. Limit theorems in the context of multivariate long-range dependence. *Stochastic Processes and their Applications*, 130(9):5394–5425, 2020: [\[doi\]](#) [\[arXiv\]](#).
- [13] M.-C. Düker and V. Pipiras. Asymptotic results for multivariate local Whittle estimation with applications. *8th IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pages 1–5, 2019: [\[doi\]](#) [\[Link\]](#) [\[Supplement\]](#).
- [14] M.-C. Düker. Limit theorems for Hilbert space-valued linear processes under long-range dependence. *Stochastic Processes and their Applications*, 128(5):1439–1465, 2018: [\[doi\]](#) [\[arXiv\]](#).

PREPRINTS

- [14] C. M. Crawford, M.-C. Düker, Y. Kim, V. Pipiras, B. L. Fredrickson, Z. F. Fisher. Analytic Standard Errors for Latent Gaussian Discrete-Valued Multivariate Time Series. *Preprint*, 2026: [\[arXiv\]](#).
- [15] M.-C. Düker and P. Zoubouloglou. The Fourth-Moment Theorem on Hilbert Spaces. *Preprint*, 2025: [\[arXiv\]](#).
- [16] M.-C. Düker and A. Waterbury. Kernel Estimation for Nonlinear Dynamics. *Preprint*, 2025: [\[arXiv\]](#).
- [17] M. Jauch, M.-C. Düker and P. Hoff. Prior distributions for structured semi-orthogonal matrices. *Preprint*, 2025: [\[arXiv\]](#).
- [18] M.-C. Düker and P. Zoubouloglou. Scaling limits for sample autocovariance operators of Hilbert space-valued linear processes. *Preprint*, 2025: [\[arXiv\]](#).
- [19] M.-C. Düker and P. Zoubouloglou. Breuer-Major Theorems for Hilbert Space-Valued Random Variables. *Preprint*, 2024: [\[arXiv\]](#).

- [20] M.-C. Düker and V. Pipiras. Testing for common structures in high-dimensional factor models. *Preprint, 2024*: [\[arXiv\]](#).

GRANTS

DFG Sachbeihilfen “Machine Learning for nonlinear dynamics” (~ 243.500 Euros)	2026
BaCaTeC-Projekt (~ 8.000 Euros)	2026
Visiting Professor Program of the Bavarian State Ministry of Science and the Arts (~ 1600 Euros)	2025
Emerging Talents Initiative Grant (~ 14.000 Euros)	2024

AWARDS AND FELLOWSHIPS

Outstanding thesis award in STEM, Dr. Heinrich-Kost-Preis Gesellschaft der Freunde der RUB	
Ph.D. Fellowship, DFG (German Research Foundation) funded, Research Training Group 2131 <i>High-dimensional Phenomena in Probability</i>	2016–2019

CERTIFICATES

Certificate for Teaching in Higher Education of the Bavarian Universities	2024
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STUDENTS

Postdoc: Panagiotis Andreou (2026–), Yuhan Tian (2025–)
 Doctoral Students: Francesco Delli Guanti (2026–), James Lavis (2025–)
 Advisor for several Master and Bachelor theses.

EMPLOYMENT

<i>Assistant Professor</i> (tenure track, W2 to W3), Technical University of Munich, Department of Mathematics	2026–present
<i>Assistant Professor</i> (tenure track, W1 to W2), Friedrich Alexander University Erlangen-Nürnberg, Department of Statistics and Data Science	2023–2026
<i>Postdoctoral Associate</i> , Cornell University, Department of Statistics and Data Science	2020–2023
<i>Research Assistant</i> , Ruhr-University Bochum, Faculty of Mathematics Research Training Group 2131 <i>High-dimensional Phenomena in Probability</i>	2016–2020
<i>Student Research Assistant</i> , Ruhr-University Bochum, Faculty of Mathematics Collaborative Research Center 823 <i>Statistical modeling of nonlinear dynamic processes</i>	2015–2016
<i>Student Teaching Assistant</i> , Ruhr-University Bochum, Faculty of Mathematics	2015–2016
<i>Student Research Assistant</i> , Ruhr-University Bochum, Institute of Hydrology, Water Resources Management and Environmental Engineering	2014–2015

TEACHING

Lecturer Special topics in Statistical learning theory (Winter term 2025/26)

Introduction to mathematical data analysis (Summer term 2025)

Advanced topics in time series (Winter term 2024/25)

Seminar in Statistical foundations of Data Science (Winter term 2024/25)

Introduction to mathematical data analysis (Summer term 2024)

Seminar in Statistical foundations of Data Science (Summer term 2024)

Lecturer Special topics in Statistical learning theory (Summer term 2023 and Winter term 2023/24)

Seminar in Statistical foundations of Data Science (Winter term 2023/24)

Teaching Assistant Financial Mathematics (Summer term 2020)

Teaching Assistant Analysis III (Winter term 2017)

Teaching Assistant Analysis I (Winter term 2016)

Co-advisor for the seminar "Statistics in everyday life" (Winter term 2015)

PROFESSIONAL SERVICE AND OUTREACH

SMSA, 16th Workshop on Stochastic Models, Statistics and Their Applications (2026).

Co-organizer: Workshop "Probability Theory and its Applications" (2025).

Chair for organizing the Section Time Series at the 17th German Probability and Statistics Days (GPSD 2025).

Session organizer: Bernoulli-IMS, 11th World Congress in Probability and Statistics (2024).

Colloquium organizer: Stochastics and Data Science Seminar at FAU Winter term 23/24.

Co-organizer KDD 2021 Workshop *Risk identification and quantification in complex human-natural systems via convergent data intensive research*.

Equal opportunities representative, Faculty of Mathematics, *Ruhr-University Bochum* (2017).

Equal opportunities representative, Department of Data Science, *FAU* (2023–2026).

Associate Editor for the Journal *Data Science in Science*.

Referee for the following journals:

Annals of Statistics	ASA Student paper award 2021
Biometrika	IEEE-Transactions on signal processing
Stochastic Processes and their Applications	IEEE-Transactions on information theory
Journal of Econometrics	Journal of time series analysis
Journal of Statistical Planning and Inference	International Statistical Review
Econometrics and Statistics	Journal of Computational and Graphical Statistics
The American Statistician	Statistica Sinica
Journal of the Korean Statistical Society	
Applications of Mathematics	

German, English

R, Python, MATLAB, C, VBA

R package development

PRESENTATIONS

BIRS, Catastrophic Events in the Complex World: Mathematics & Statistics of extremes in the Age of Machine, Banff, August 2026 (invited).

GOFCP, Workshop on Goodness-of-fit, Change-point and Related Problems, Prague, August 2025 (invited).

Joint Statistical Meetings (JSM), Nashville (TN), August 2025 (invited).

Workshop on Functional Data Analysis, Bonn, June 2025 (invited).

Seminar, Statistical Laboratory, University of Cambridge, January 2025 (invited).

Seminar, Department of Statistical Sciences, Duke University, January 2025 (invited).

18th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, 14–16th December 2024 (invited).

Seminar, Department of Mathematics, Otto-von-Guericke University Magdeburg, November 2024 (invited).

7th International Conference on Econometrics and Statistics (EcoSta 2024), Beijing, 17–19th July 2024 (invited).

16th International Conference of the ERCIM WG on Computational and Methodological Statistics, Berlin, 16–18th December 2023 (invited).

Seminar, Department of Mathematics, University of Athens, December 2023 (invited).

Seminar, Department of Mathematics, EPFL, November 2023 (invited).

NBER-NSF Time Series Conference, Montreal, September 2023.

Joint Statistical Meetings (JSM), Toronto, August 2023 (invited).

6th International Conference on Econometrics and Statistics (EcoSta 2023), Waseda University Tokyo, virtual, August 2023 (invited).

13th Extreme Value Analysis Conference, Bocconi University Milan, June 2023 (invited).

Conference “Women in Data Science”, Friedrich-Alexander University Erlangen, virtual, April 2023 (invited).

Seminar, Department of Statistics and Mathematics, University of Massachusetts at Amherst, March 2023 (invited).

Conference “Adaptive and high-dimensional spatio-temporal methods for forecasting”, CIRM Luminy, September 2022 (invited).

Seminar, Econometrics and Statistics, University of Chicago, Booth School of Business, virtual, March 2022 (invited).

Seminar, Department of Statistics, University of Wisconsin Madison, virtual, February 2022 (invited).

Seminar, Department of Statistics, University of Michigan, January 2022 (invited).

10th International Conference of the ERCIM WG on Computational and Methodological Statistics, virtual, 18–20th December 2021 (invited).

Stochastic Seminar, Department of Mathematics, University of Utah, October 2021 (invited).

Joint Statistical Meeting 2021, virtual, 7–12th August 2021.

10th World Congress, virtual, July 19th–23rd.

Graduate Seminar, Department of Statistics and Operations Research, UNC Chapel Hill, September 2019.

40th Conference on Stochastic Processes and their Applications, Gothenburg, 11–15th of June 2018.

13th German Probability and Statistics Days, Freiburg, 27th February–02nd March 2018.

10th International Conference of the ERCIM WG on Computational and Methodological Statistics, London, 16–18th December 2017.

Graduate Seminar, Department of Statistics and Operations Research, TU Dortmund, November 2017.

31st European Meeting of Statisticians, Helsinki, 24–28th July 2017.

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